Chitrarth Rai

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Professional Summary

Full-Stack & Systems Software Engineer specializing in distributed backend infrastructure, high-availability storage, low-latency microservices, and native mobile architectures. Experienced across Kubernetes (k3s), CloudNativePG, MinIO S3, FastAPI, Node.js, React Native (Kotlin JNI), and PostgreSQL/MongoDB optimizations. Proven track record deploying zero-drift disaster recovery meshes, 60 FPS camera pipelines, enterprise VAPT hardening, and automated CI/CD pipelines.

Technical Skills

Languages	TypeScript, JavaScript (ES6+, Async/Await), Python, Kotlin, C++, SQL, Bash, HTML/CSS
Frontend & Mobile	React.js, Next.js 14, React Native, Expo, Reanimated, Redux Toolkit, Zustand, React Query, Tailwind CSS, Three.js
Backend & Distributed	FastAPI, Node.js, Express.js, CloudNativePG (HA, PITR), Apache Kafka, BullMQ, Celery, REST/JSON-RPC, Swagger/OpenAPI
Cloud, DevOps & DR	Kubernetes (k3s), Ansible, OpenTofu, Docker, WireGuard VPN, Envoy/Kong Gateway, Azure, Supabase, MinIO S3 (WORM)
Testing, AI & Security	pytest, Jest, Cypress, Postman, ONNX Runtime, OpenCV, Gemini/OpenAI API, LangChain, RAG, WebRTC, WebSockets, VAPT, CI/CD

Work Experience

Neophyte AI *Feb 2025 – Present*
Software Engineer *Navi Mumbai, India*

- **High-Performance Edge AI & Native Camera Engine:** Architected an asynchronous edge-compute camera pipeline in **React Native** utilizing a custom **Kotlin Native Module** with direct **JNI zero-copy buffer manipulation** and **ONNX Runtime**; eliminated JavaScript bridge serialization bottlenecks to sustain **60 FPS** under continuous 12MP capture, cutting inference latency by **40%**.
- **Distributed HA Storage & Disaster Recovery (Neometry):** Designed a multi-region, zero-egress storage mesh on **Kubernetes (k3s)** orchestrating multi-node **MinIO S3** clusters and **CloudNativePG**; implemented automated failover, Point-In-Time Recovery (PITR), and encrypted async replication over **WireGuard VPN** via **Ansible/OpenTofu**, enforcing SEC Rule 17a-4 compliance with **15-day immutable WORM locking**.
- **Distributed Microservices & High-Throughput Data Layer:** Engineered distributed REST and JSON-RPC services with async **FastAPI** (Pydantic, **Swagger/OpenAPI contracts**, OAuth2/JWT) and **Node.js**; deployed an in-memory **Redis caching layer** with proactive cache invalidation, connection pooling, and payload compression, slashing p99 API latency by **55%** under high-concurrency production workloads.
- **Enterprise Retail Analytics & Query Optimization:** Spearheaded an analytics platform in **React 18/TypeScript** (React Query) leveraging DOM windowing/virtualization to render 50,000+ real-time telemetry points at **60 FPS**; analyzed **PostgreSQL query plans** (EXPLAIN ANALYZE) and re-engineered multistage **MongoDB Aggregation Pipelines** with compound indexes across 2M+ SKU records, slashing database query execution time by **60%**.
- **Offline-First Mobile Architecture & Automated CV Engines:** Engineered a resilient offline sync engine with transactional **SQLite caching** and background reconciliation (45% faster TTI); developed headless **OpenCV** image processing workers managing a daily queue of **1,500+ audit jobs** for automated planogram and packaging compliance validation, accelerating retail audit turnaround by **70%**.
- **Enterprise VAPT Hardening & DevOps Automation:** Hardened core mobile and cloud infrastructures for enterprise production deployment (Reliance) through dynamic **SSL certificate pinning**, AES-256/RSA encryption, and ProGuard bytecode obfuscation; architected automated CI/CD pipelines in **GitHub Actions** running **pytest and Jest test suites** on every PR before zero-downtime container rollouts.

Technical Projects

ChatX – Enterprise Collaboration Platform GitHub

- **Technologies:** Next.js 14, React Native (Expo), TypeScript, Tailwind CSS, Supabase (Realtime, Edge Functions), WebRTC, WebSockets, Zustand.
- Architected a real-time messaging and WebRTC video conferencing platform supporting sub-100ms multi-client state sync for 3-tier message delivery status (sent, delivered, read), dynamic user presence, and live WebSocket telemetry feeds.
- Enforced enterprise security with granular PostgreSQL Row-Level Security (RLS) policies, session-based PKCE auth, channel PIN-locking, and serverless TypeScript Edge Functions; automated PR validation with GitHub Actions CI/CD.

FinanceTask – AI-Powered Financial Tracker Website | Application | GitHub

- **Technologies:** React.js, React Native, TypeScript, Supabase Realtime, PostgreSQL, Gemini AI, Kotlin Native Module, Recharts.
- Built a cross-platform finance manager syncing state across web/mobile via Supabase WebSockets and PostgreSQL triggers with sub-100ms latency; styled responsive dashboard interfaces with WCAG-compliant contrast.

– Engineered an Android Kotlin Native Module for offline background SMS scraping, utilizing Gemini AI zero-shot NER to parse bank transactions and extract deadlines with 95% accuracy; enforced automated Jest test coverage.

FedEx Document Processing Pipeline GitHub

– **Technologies:** Python, OpenCV, Tesseract OCR, Multiprocessing, JSON Schema Validation, OpenAI API.

– Developed an automated OCR parsing pipeline using OpenCV (perspective correction, adaptive thresholding) and Tesseract to extract shipping invoice data, reducing manual processing time by 90%.

– Implemented a multi-threaded Python worker pool to concurrently batch-process multi-page PDFs with schema validation and automated pytest coverage, increasing processing throughput by 75%.

Education

IIIT Bhubaneswar *2021 – 2025*

B.Tech in Electronics & Telecommunication Engineering *Bhubaneswar, India*

Leadership & Interests

Core Interests: Distributed Systems, High-Performance Mobile Architectures, Computer Vision, System Design.

Campus Leadership: Active member of Film & Theatrical Society (FATS) and Photogeeks Society, IIIT Bhubaneswar.